
Customer Churn Prediction

Business Brief | ML Capstone Project

Prepared for	Domain	Methodology	Model
Customer Experience (CXM) Practice	Churn Analytics	Supervised Binary Classification	LightGBM — Gradient Boosted Trees

01 | BUSINESS OBJECTIVE

Customer attrition is one of the most measurable levers of revenue loss. Research consistently shows that acquiring a new customer costs five to seven times more than retaining an existing one. Despite this, most retention budgets are spent reactively — after a customer has already disengaged.

This project builds a proactive churn prediction system that identifies customers who are at high risk of leaving before they disengage. By surfacing risk scores ahead of time, the business can redirect retention budget toward the customers most likely to respond to intervention.

The model specifically supports two business decisions:

- Which customers should receive a targeted retention offer in the next campaign cycle?
- How should the retention team prioritise its limited outreach capacity?

02 | DATA & TARGET VARIABLE

Four CSV files were provided — two for training and two for test — covering customer profiles and behavioural signals. They were merged on customer_id to produce one row per customer.

Dataset	Records	Columns	Notes
train_customer_profile	15,000	12	Demographics, contract, loyalty — includes churn label
train_customer_behavior	15,000	14	Transactions, engagement, support interactions
test_customer_profile	5,000	11	Same schema, no churn label
test_customer_behavior	5,000	14	Same schema, no churn label

Target Variable: churn (binary integer). A value of 1 indicates the customer churned; 0 indicates they were retained. The positive class (churn = 1) is the minority class, representing approximately 21.3% of training customers — a 3.71:1 imbalance ratio that required explicit handling.

03 | FEATURE & DATA PREPARATION

Data Cleaning

The raw data contained several quality issues identified during EDA. All cleaning was applied identically to both train and test sets to prevent leakage.

Issue	Affected Columns	Resolution
Inconsistent casing & abbreviations	gender, region, contract_type, marital_status	Fuzzy matching (RapidFuzz ≥ 90% similarity) + canonical mapping
Redundant category labels	contract_type ('multi-year', 'multiyear', 'Multi year')	Merged into single canonical label 'Multi-Year'
Derived-column nulls	purchase_frequency (~10% missing)	Imputed as total_spend_6m / avg_order_value
Outliers	avg_order_value, total_spend_6m, recency_days, app_sessions, support_tickets	Clipped to 1st–99th percentile
Impossible values	returns_ratio	Clipped to [0, 1]

Feature Engineering — 22 New Features

All features were engineered using only information available at prediction time. No future-looking signals were introduced. The engineered set is grouped into five families:

Feature Family	Key Features	Business Rationale
Customer Value	spend_per_purchase, monthly_customer_value, spend_per_category	Captures economic importance and depth of product engagement
Engagement	engagement_score (email + campaign + app), purchase_conversion_efficiency	Composite signals of customer relationship health
Risk / Friction	dissatisfaction_score, support_ticket_rate, inactivity_score	Proxy for negative service experiences
Loyalty & Contract	loyalty_rank, monthly_contract_flag, loyalty_contract_risk, tenure_loyalty_gap	Short-term contracts and low loyalty significantly predict churn

Feature Family	Key Features	Business Rationale
Behavioural Flags	high_recency_flag, low_engagement_flag, high_return_flag, high_support_flag	Binary triggers for threshold-based risk criteria
Segmentation Buckets	tenure_bucket, recency_bucket, spending_bucket, age_group	Ordinal groupings for non-linear interactions
Customer Cluster	cluster (K-Means, K=4, RFM-inspired)	Unsupervised segment label capturing customer behaviour archetypes

Preprocessing Pipeline

A scikit-learn ColumnTransformer pipeline was fitted exclusively on training data and then applied to the validation and test sets without re-fitting. Numeric features were imputed with the median and scaled with StandardScaler. Categorical features were imputed with the mode and encoded with OrdinalEncoder (unknown values mapped to -1).

04 | MODEL SELECTION

Six candidate models were trained and evaluated under two resampling strategies (SMOTE oversampling and Tomek Links undersampling) — twelve configurations in total. LightGBM trained on Tomek-resampled data was selected as the final model.

Model	Resampling	ROC-AUC	Accuracy	F1-Score	Avg Precision
LightGBM	SMOTE	0.7718	0.8333	0.5088	0.5707
Gradient Boosting	SMOTE	0.7625	0.8340	0.4990	0.5632
Random Forest	SMOTE	0.7416	0.8267	0.5149	0.5249
XGBoost	SMOTE	0.7306	0.7730	0.4837	0.5098
Logistic Regression	SMOTE	0.7210	0.6680	0.4639	0.4556
Decision Tree	SMOTE	0.5983	0.7107	0.3719	0.2661
Gradient Boosting	Tomek	0.7701	0.8333	0.4960	0.5683
LightGBM	Tomek	0.7687	0.7923	0.5585	0.5805
Random Forest	Tomek	0.7634	0.8253	0.4165	0.5536
XGBoost	Tomek	0.7570	0.8010	0.5439	0.5493
Logistic Regression	Tomek	0.7223	0.6613	0.4636	0.4578
Decision Tree	Tomek	0.6089	0.7300	0.3854	0.2767

Model	Resampling	ROC-AUC	Accuracy	F1-Score	Avg Precision
LightGBM (tuned) ✓	Tomek	0.7745	0.7967	0.5479	0.5849

LightGBM was selected for the following reasons:

- Highest ROC-AUC (0.7745) and F1-Score (0.5479) on the held-out validation set.
- Native handling of missing values removes dependency on imputation quality for tree splits.
- Gradient boosting captures complex non-linear interactions between contract type, engagement, and tenure that linear models miss.
- Computationally efficient with `class_weight='balanced'`, making it production-suitable at scale.
- Tomek Links (borderline noise removal) consistently outperformed SMOTE oversampling across all boosted models by producing sharper decision boundaries.

Hyperparameter Tuning

RandomizedSearchCV with 60 candidates and 5-fold stratified cross-validation was used to tune nine hyperparameters including learning rate (0.005), number of estimators (748), `num_leaves` (81), and `max_depth` (10). Optimisation target was ROC-AUC. Best cross-validated ROC-AUC prior to final refit: 0.7712.

05 | METRIC & BUSINESS TRADE-OFF

0.774
ROC-AUC

0.548
Macro F1-Score

ROC-AUC was the primary optimisation metric during model selection and hyperparameter tuning. It measures the model's ability to rank churners above non-churners across all possible thresholds, making it the most robust metric under class imbalance.

Macro F1-Score (equally weighting both classes) was used as the secondary business metric because the evaluation criteria require predictive performance across both retained and churned customers, not just the majority class.

Precision vs. Recall Trade-off

Metric	Retained (0)	Churned (1)	Business Interpretation
Precision	89%	50%	When we flag a churner, we are right 50% of the time
Recall	84%	60%	We capture 60% of all customers who will actually churn
F1-Score	0.86	0.55	Harmonic mean, penalising both false positives and misses

In a retention context, missing a churner (false negative) is typically more costly than an unnecessary outreach (false positive). The current threshold of 0.5 can be lowered to increase recall at the cost of more false alarms — the appropriate threshold should be calibrated to the unit economics of the retention offer versus the lifetime value at risk.

06 | VALIDATION EVIDENCE

Model performance was evaluated through a rigorous three-layer validation framework to guard against overfitting and data leakage.

Validation Layer	Description	Result
Hold-out validation set	20% stratified split (3,000 customers) held back from all training and preprocessing fitting	ROC-AUC 0.7745 F1 0.548
5-Fold Stratified CV (tuning)	Cross-validation on Tomek-resampled training data during hyperparameter search	Best CV ROC-AUC 0.7712
5-Fold CV — Final check	Applied to full training data (all 15,000 rows) resampled with SMOTE, using best hyperparameters	ROC-AUC 0.9415 ± 0.0039

The high 5-fold CV score on the full resampled training set (0.94) reflects the model learning from balanced, complete data. The more conservative hold-out score (0.77) is a better estimate of real-world performance on unseen customers drawn from the natural 21/79 class distribution.

Baseline Comparison

Model	ROC-AUC	Accuracy	F1-Score	Avg Precision
Logistic Regression — SMOTE (balanced)	0.721	0.668	0.464	0.456
LightGBM — Tuned Final Model	0.7745	0.7893	0.5479	0.5849

The final model improves ROC-AUC by 27.4 percentage points and Avg Precision by 37.2 percentage points over the naive baseline, and outperforms the simplest interpretable model (Logistic Regression) by a meaningful margin across all three metrics.

07 | KEY CHURN DRIVERS

LightGBM's feature importance (split-count method) reveals the most influential predictors. These findings should inform the design of retention interventions.

Rank	Feature	Type	Implication for Retention Teams
1	customer_satisfaction_score	Profile	Low satisfaction is the single strongest churn signal — CSAT recovery programmes should be prioritised
2	dissatisfaction_score (engineered)	Engineered	Composite of returns, support tickets, and low satisfaction — flags customers in friction loops
3	purchase_frequency	Behaviour	Declining purchase cadence precedes churn — trigger re-engagement at first sign of drop
4	campaign_response_rate	Behaviour	Non-responsive customers are at risk — low response to campaigns warrants channel experiment
5	engagement_score (engineered)	Engineered	Holistic email + app + campaign signal — low engagement customers need multi-touch outreach
6	email_open_rate	Behaviour	Direct inbox engagement; customers who stop opening emails are drifting away
7	recency_days	Behaviour	Days since last purchase — strong recency predictor, supports RFM-based urgency scoring
8	returns_ratio	Behaviour	High return rates suggest product-fit mismatch — merchandise or recommendation intervention needed
9	contract_type	Profile	Monthly contract customers churn at significantly higher rates than annual or multi-year customers

08 | BUSINESS RECOMMENDATIONS

How to Use the Model Output

The model produces a churn probability score (0–1) for each customer. Retention teams should use this score to prioritise outreach rather than applying a single binary cut-off. A suggested three-tier action framework:

Risk Tier	Score Range	Recommended Action	Owner
High Risk	≥ 0.55	Personalised outreach within 7 days — loyalty upgrade offer, dedicated account manager call	Retention team
Medium Risk	0.35 – 0.54	Automated multi-channel nurture sequence — targeted email + in-app message + next-best offer	CRM / Marketing automation
Low Risk	< 0.35	Standard engagement programme — no special intervention required	Business as usual

Specific Intervention Recommendations

- Contract upgrade incentives — offer Monthly customers a discounted Annual plan. Monthly customers exhibit the highest churn rate; locking them into longer contracts is the single highest-ROI structural intervention.
- Satisfaction recovery programme — customers with CSAT below 40 and a dissatisfaction_score in the top quartile should receive a proactive service recovery call within 48 hours of the score being recorded.
- Re-engagement triggers — customers who have not purchased in more than 90 days (high_recency_flag = 1) should automatically enter a win-back sequence starting with a personalised best-seller recommendation.
- Returns-driven merchandise review — customers with returns_ratio above 0.30 may be experiencing chronic product-fit issues. Flag these accounts for a category recommendation review or personal stylist / advisor programme.
- Cluster-targeted campaigns — Cluster 0 (churn rate 26.6%) should be the priority segment; Cluster 2 (churn rate 13.7%) can receive lower-intensity engagement to maintain their healthy status.

09 | LIMITATIONS & ASSUMPTIONS

Limitation	Detail & Mitigation
Temporal scope	All behavioural features are aggregated over a 6-month window. The model does not capture seasonal patterns or long-term trend signals. Mitigation: retrain quarterly with fresh data.
Label definition	The definition of 'churned' is not documented in the dataset. If churn is defined as 90-day inactivity vs. contract cancellation, the model's applicability changes. Clarify with the client before deployment.
Hold-out performance vs. live	Validation ROC-AUC (0.77) was measured on a random hold-out. In a live deployment, customers may be scored at different time offsets, which could create distribution shift. Champion-challenger testing is recommended.
Class imbalance	The 3.71:1 imbalance was handled via Tomek Links and class weighting. The model still has limited recall (60%) for the minority class. Threshold tuning against business unit economics can improve this.
Feature availability	Some features (e.g. last_promotion_used, loyalty_tier) had 10–15% missing values. While median/mode imputation was used, systematic missingness in a production pipeline should be monitored.
Causal inference	The model identifies correlation, not causation. A high churn score does not guarantee intervention will succeed — treatment effect estimation (e.g. uplift modelling) is a recommended next step for advanced ROI measurement.
